

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	17 July 2026
Team ID	-----
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Used to develop a responsive and user-friendly web interface for entering weather parameters and displaying flood prediction results. Bootstrap ensures responsive design across different devices.
2	Backend / Server-Side	Python, Flask Framework	Python provides powerful machine learning capabilities, while Flask offers a lightweight framework for integrating the trained model with a web application and handling prediction requests efficiently.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
3	Database / Data Storage	CSV Dataset (Historical Flood Dataset)	The application uses historical flood and rainfall data stored in CSV format for model training, testing, and prediction. This approach is simple, lightweight, and sufficient for the project scope.
4	Cloud / Hosting / Deployment	IBM Cloud / Local Flask Server	IBM Cloud supports AI application deployment and scalability, while the Flask development server enables local testing during development. This aligns with the IBM SkillsBuild internship environment.
5	Version Control & CI/CD	Git, GitHub	Git is used for source code management, while GitHub provides version control, collaboration, project backup, and repository hosting throughout the development lifecycle.
6	Third-Party APIs / Other Tools	XGBoost, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib, Jupyter Notebook, VS Code	XGBoost is used for flood prediction. Scikit-learn supports preprocessing and evaluation. Pandas and NumPy handle data manipulation. Matplotlib and Seaborn are used for EDA and visualization. Joblib stores the trained model. Jupyter Notebook is used for model development, and VS Code is used for application development.

Additional Development Tools (Optional)

Category	Technology	Purpose
Machine Learning Algorithm	XGBoost	Predict flood occurrence with high accuracy
Data Processing	Pandas, NumPy	Data cleaning, preprocessing, and manipulation
Visualization	Matplotlib,	Exploratory Data Analysis (EDA) and graphical visualization

Category	Technology	Purpose
	Seaborn	
Model Evaluation	Scikit-learn	Train-test split, performance metrics, confusion matrix, classification report
Model Serialization	Joblib	Save and load the trained XGBoost model for deployment
IDE	Visual Studio Code	Develop and debug the Flask application
Notebook	Jupyter Notebook	Data analysis, experimentation, and model training